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INSTITUTE OF PUBLIC
ADMINISTRATION

Certificate in Local Government Studies
(Accredited by NUI)

LESSON 8
Climate Change

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LEARNING OBJECTIVES:

By the end of this lesson you should:

- Understand the developing role of local government in the area of climate change, and initiatives at local level as Ireland moves towards a low carbon and climate resilient society.
- Appreciate links between climate change and the work of local authorities in a wide range of other service areas.

8.0 INTRODUCTION AND OVERVIEW

This lesson details local government's evolving role in the field of climate change, arguably the central challenge facing society now and for the foreseeable future. Climate change is a theme that has implications for the planet, for Ireland, and for local areas. The Environmental Protection Agency (EPA) website notes the scale of the challenge:

> Climate change is one of the most significant and challenging issues currently facing humanity. Increased levels of greenhouse gases, such as CO_2, increase the amount of energy trapped in the atmosphere which leads to global impacts such as increased temperatures, melting of snow and ice and rising global average sea-level.
>
> If not addressed, the projected impacts of climate change present a very serious risk of dangerous and effectively irreversible climate impacts at both the global and national levels with food production and ecosystems being particularly vulnerable. Ireland also faces a range of negative impacts related to climate change. Consequently, future investment decisions at national, regional and local levels must incorporate consideration of future climate conditions and adaptation options, with flood prevention and improved management of water resources being important elements of these considerations.

Climate change has been the subject of negotiations at global (UN) and European Union level, but many of the implications and actions needed to cope with a changing climate are needed at local level – hence the maxim sometimes advanced around climate change: "Think Globally, Act Locally".

What this means of course is that climate change has implications for all of us, as individuals, as communities, and as organisations. From a local government perspective, it has implications across the range of different service areas. While originally there might have been an instinctive link made between climate change and environmental protection, it is becoming increasingly obvious that there are links between climate change and many other local government services: consider for example the climate change dimension to planning, the climate change dimension to housing and building, the climate change dimension to roads and transportation, the climate change dimension to economic and local development, the climate change dimension to emergency planning, etc.

Therefore, at the outset it should be stressed that this lesson should not be read in isolation. Rather students should also use this Lesson to consider the range of implications and identify linkages with other service areas. We start with a discussion of what is meant by sustainable development, international agreements on climate change, national policy in the area, before turning to local government's evolving role in climate change.

8.1 SUSTAINABLE DEVELOPMENT

The concept of sustainable development was given international recognition at the UN Conference on the Environment and Development, which was held in Rio de Janeiro in 1992, and now commonly referred to as the "Earth Summit". Sustainable development is commonly defined as "*development which meets the needs of the present, without compromising the ability of future generations to meet their own needs"* (World Commission on Environment and Development, 1987, p. 43)

The Earth Summit produced <u>five</u> documents:

1. The Rio Declaration on Environment and Development – 27 principles defining the rights and responsibilities of nations as they pursue human development and well-being.
2. Agenda 21– a blueprint on how to make development socially, economically and environmentally sustainable in the 21st century.
3. A Statement of Principles to guide the sustainable development of forests.
4. A Framework Convention on Climate Change to stabilise greenhouse gases.
5. A Convention on biological diversity.

While each of these is important, we will focus on two below – namely Agenda 21 and the Convention on Climate Change.

These developments were also followed up by the adoption in 2015 of the 2030 Agenda on Sustainable Development and the UN Sustainable Development Goals (SDGs). The 17 SDGs span different dimensions of sustainable development, encompassing not only goals around environmental protection but also economic development and social progress. These include goals around poverty and inequality, hunger, health and well-being, education, gender equality, employment and economic growth, as well as responsible consumption and production, climate action, nature, sustainable cities and communities, and access to water, energy and infrastructure.

8.1.1 AGENDA 21

Agenda 21 is the United Nations' Action Plan to achieve global sustainable development. It is a far-reaching statement on the basis of which world leaders committed themselves to building a more sustainable world. Some of the major aims outlined in Agenda 21 include:

- reducing the amount of energy and raw materials society consumes, as well as the pollution and waste it produces
- protecting fragile ecosystems and environments
- bringing about a fairer distribution of wealth, both between countries and between different social groups within countries, with particular emphasis on the rights of poor and disadvantaged people.

Local Agenda 21

Agenda 21 called on local authorities throughout the world to set out their 'local agenda' to achieve sustainable development at local level. The initiatives developed at the local authority level comprised a Local Agenda 21 plan.

Chapter 28 of Agenda 21 stated that:

> "Local Authorities construct, operate and maintain economic, social and environmental infrastructure, oversee planning processes, establish local and environmental policies and regulations, and assist in implementing national and sub-national policies. As the level of government closest to the people, they play a vital role in educating, mobilising and responding to the public to promote sustainable development."

Although progress by local authorities on Local Agenda 21 was slow, the establishment of public participation networks and the move towards a more partnership and participatory approach to decision-making under both "Better Local Government" and "Putting People First" can be seen as a positive step in the Agenda 21 direction.

8.1.2 Climate Change – International Commitments

In the 1980s and early 1990s scientific evidence linking greenhouse gas emissions from human activities to climate change gave rise for concern. The Intergovernmental Panel on Climate Change (IPCC) was established in 1988 to provide decision-makers and others interested in climate change with an objective source of information about climate change. It is a scientific intergovernmental body set up by the World Meteorological Organisation and the United Nations Environment Programme. The findings of the first IPCC Assessment Report played a decisive role in leading to the United Nations Framework Convention on Climate Change (UNFCCC) (1992). This entered into force in 1994 and provided the overall policy framework for addressing the climate change issue with the objective of preventing dangerous man-made interference with climate change. However, the Convention did not identify what constituted dangerous climate change or how it might be prevented, and commitments to reduce greenhouse gas emissions were not legally binding. The voluntary efforts arising proved to be inadequate.

The IPCC Second Assessment Report, 1995 provided key input into the negotiations of the Kyoto Protocol in 1997. The Kyoto Protocol, which came into force in 2005, provided for individual legally binding targets to be achieved by industrialised or developed countries.

Ireland agreed, under an EU burden-sharing agreement, to a target limiting its greenhouse gas emissions to 13% above the 1990 levels by the first commitment period as part of its contribution to the overall EU target.

The Paris Agreement was adopted in Paris in 2015. Countries signing the agreement agreed to limit the global temperature increase to 2°C while pursuing efforts to limit it to 1.5°C. This was the first-ever universal, legally binding global climate change agreement, and provides a framework to reduce emissions. The agreement also recognises the role of cities and local authorities in reducing emissions, and in promoting cooperation on a regional and international level.

Subsequently, the European Union adopted the objective of achieving carbon neutrality in the EU by 2050 by transitioning to a clean, circular economy as well as restoring biodiversity (known as the 'European Green Deal'). The European Climate Law was adopted in 2021 (Regulation 2021/1119) to make the targets in the UN Framework Convention on Climate Change (UNFCCC) and the European Green Deal legally binding for EU states including Ireland. This includes achieving carbon neutrality by 2050, and an earlier target of reducing net greenhouse gas emissions by 55% by 2030 (compared to 1990 levels). The EU's programme identifies where investments will be required as well as ensuring how an inclusive and just transition can be achieved. Delivering on this commitment will require the transformation of a range of sectors including energy, industry, agriculture, transport and the built environment.

8.2 CLIMATE CHANGE PROJECTIONS FOR IRELAND

Long-term climate projections for Ireland indicate the prospect of the following changes:

- Increase in temperatures.
- Increased number of heatwaves and a possibility of drier summers, especially in the east.
- Increased precipitation and rainfall in the winter, especially in the west, with an increase in the amount of water flowing in rivers, thus increasing the risk of flooding.
- While slightly fewer storms are projected, these are expected to be more intense.

From a local government perspective, one can anticipate the following potential implications:

- Increased rate of deterioration of roads, footpaths, cycleways, bridges and other transport infrastructure from flooding events, sea level rise, storm surges, sustained higher temperatures (or a combination of these).
- Increased damage to local authority buildings and housing stock.
- Existing flood defences being insufficient to cope with higher rainfall events.
- Deterioration of community infrastructure (i.e. public realm spaces, playgrounds, public parks and beaches) from sustained extreme events and coastal erosion.

- Damage to properties, streetscapes and community assets.
- Increased demands on emergency services and responders.

8.3 NATIONAL CLIMATE CHANGE STRATEGIES

Ireland published its first *National Climate Change Strategy* in 2000. This set out a series of measures in areas such as taxation, energy, transport, the industrial and services sector, agriculture, forestry and the built environment and residential sector.

The National Development Plan, 2018-2027 recognised that climate change is, by a considerable margin, the most pressing international environmental issue that we face. A national strategic outcome of the plan is to move Ireland to being a low-carbon and climate resilient society.

The Climate Action and Local Carbon Development (Amendment) Act 2021 commits Ireland to reducing its emission levels by 51% by 2030 (based on 2018 levels) and in line with EU policy objectives to becoming climate neutral by 2050.

8.3.1 National Climate Action Plan

The National Climate Action Plan was initially published in 2019 with annual updates published thereafter. The Plan outlines key issues across different sectors, including energy, transport, built environment, industry and agriculture and sets out how decarbonisation targets can be achieved. The Plan also sets out governance arrangements, including carbon-proofing our policies, establishment of carbon budgets, a strengthened Climate Change Advisory Council and greater accountability to the Oireachtas. Local government is identified as having a key role in implementing measures to achieve national climate change targets through local authority climate action plans, integrating climate change in key policies such as development plans, identifying decarbonising zones, and 'leading by example' through reducing the local authority's own carbon footprint in terms of energy use, buildings, infrastructure, vehicles, etc.

The Plan recognises that Ireland must significantly step up its commitments to tackle climate disruption. The leadership role both the Government and public bodies can play in taking early action on climate is fundamental to achieving our decarbonisation goals. Taking decisive action to confront climate disruption will be a major challenge to every dimension of our society, but the benefits are huge – warmer homes, cleaner air, a sustainable use of the world's scarce resources, more connected communities, authentic values, and quality jobs in enterprises which can compete in a decarbonised world. From Government to businesses, communities to householders, climate action is collective action.

As noted, the Plan identified local government as a key stakeholder in many actions and highlighted the leadership role that it must play in the transition to a net-zero carbon society. The plan tasks local government, along with reducing emissions from their own

operations, with becoming a key mobiliser at local and community level as well as becoming a key driver and enabler of local decarbonisation efforts.

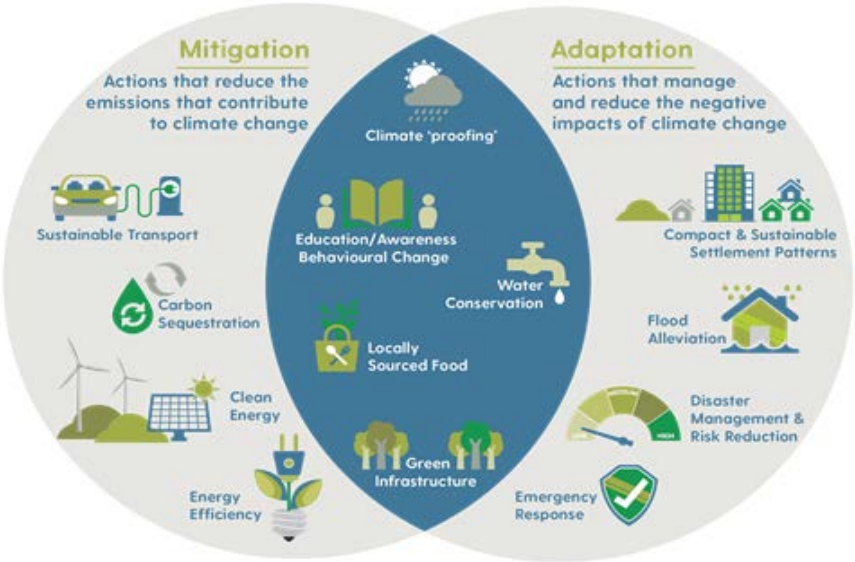
8.3.2 Mitigation and Adaptation

Two phrases often used in association with climate change are 'mitigation' and 'adaptation'. The phrase 'mitigation' refers to initiatives designed to make the impacts of climate change less severe. This can include reducing the amount of greenhouse gas emissions with a view to keeping temperature rises to a minimum.

On the other hand, 'adaptation' refers to preparing for the harmful effects of climate change such as more intense weather events, a rise in sea levels or food insecurity. The European Environment Agency defines 'adaptation' as "anticipating the adverse effects of climate change and taking appropriate action to prevent or minimise the damage they can cause or taking advantage of opportunities that may arise".

In summary, 'adaptation' suggests that some effects of climate change are inevitable and that these will become more severe at least over the next decades. Accordingly, it will be important for local communities to become more resilient as an immediate task in reducing their exposure or vulnerability to these effects as far as possible. 'Mitigation' refers to measures designed to reduce carbon emissions and perhaps be in a position at some stage to reverse the trend. Some measures, as can be noted from the diagram below, can be common to both interventions.

Figure 8.1 Mitigation and Adaptation



Source: Climate Action Regional Office

8.3.3 National Mitigation Plan

Launched by the Department of Communications, Climate Action and Environment in 2017, this cross-departmental plan seeks to accelerate the decarbonisation process in Ireland. A number of key areas are targeted by the Plan for decarbonisation, namely: electricity generation, built environment, transport, agriculture and forestry sectors.

8.3.4 National Climate Change Adaptation Framework

In 2018 the Department of Communications, Climate Action and Environment published the 'National Adaptation Framework: Planning for A Climate Resilient Ireland', which builds on an earlier 2012 Climate Change Adaptation Framework. The Framework provides the policy context for a strategic national adaptation response to climate change in Ireland setting out measures to reduce the vulnerability of areas to the negative effects of climate change.

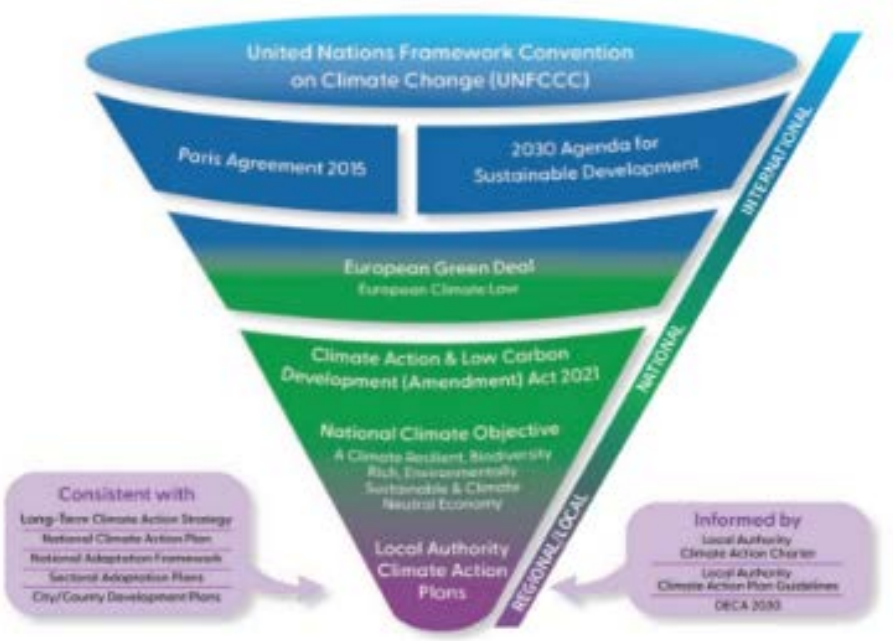
Under the Framework, the relevant Government departments and agencies are asked to commence the preparation of sectoral adaptation plans with work to commence in 2018. Local authorities were required to prepare local adaptation strategies, the first of which were prepared in 2019. The Framework was reviewed in 2022 which noted how local authorities had been proactive in developing adaptation policies and actions.

8.4 LOCAL GOVERNMENT AND CLIMATE ACTION

The Local Authority Climate Action Charter was signed by Government and every local authority in October 2019. Its aim was to ensure local authorities embed decarbonisation, climate resilience and sustainable development into every function. It tasks local authorities with delivering on energy reduction targets, carbon proofing major projects and programmes, partnering and collaborating with communities as well as building local citizen engagement.

Section 16 of the Climate Action and Low Carbon Development (Amendment) Act 2021 made it a statutory requirement that each local authority should prepare a local authority climate action plan for a 5-year period, setting out each local authority's priorities with regard to both mitigation and adaptation measures in the area of climate change. These plans are expected to reference national climate objectives. Guidelines on these plans were published by the Department of the Environment in 2023. The Figure below illustrates how local climate action plans are informed by national, EU and international frameworks.

Figure 8.2: National and International Legislative and Policy Context for Local Climate Action Plans



Source: Climate Action Regional Office

Many local authorities have established a dedicated SPC with responsibility for Climate Action. While this can provide a focus for climate action initiatives at local level, a risk can be that this area of work is seen as being exclusively based with this committee, rather than being an issue that affects all service areas. While the Guidelines on SPCs published in 2024 did not specify that that all local authorities necessarily had to have a Climate Action SPC, they did provide that

> … each SPC is required to consider climate action and climate impacts as part of any and all policies that form part of its work programme … Climate action should be considered in its widest sense, to include biodiversity action, water quality action, sustainable land use, energy (source and usage), resource management, sustainable transport / mobility, air quality / emissions, soil quality, waste management, community action etc., and other related issues – reflecting the complex and integrated nature of climate action. This should not be read as precluding a local authority from establishing a specific Environment / Climate Action / Water Quality etc. SPC if it deems it appropriate to do so (*Corporate Policy Groups and Strategic Policy Committees – Revised Guidelines for Establishment and Operation*, 2024, p. 21).

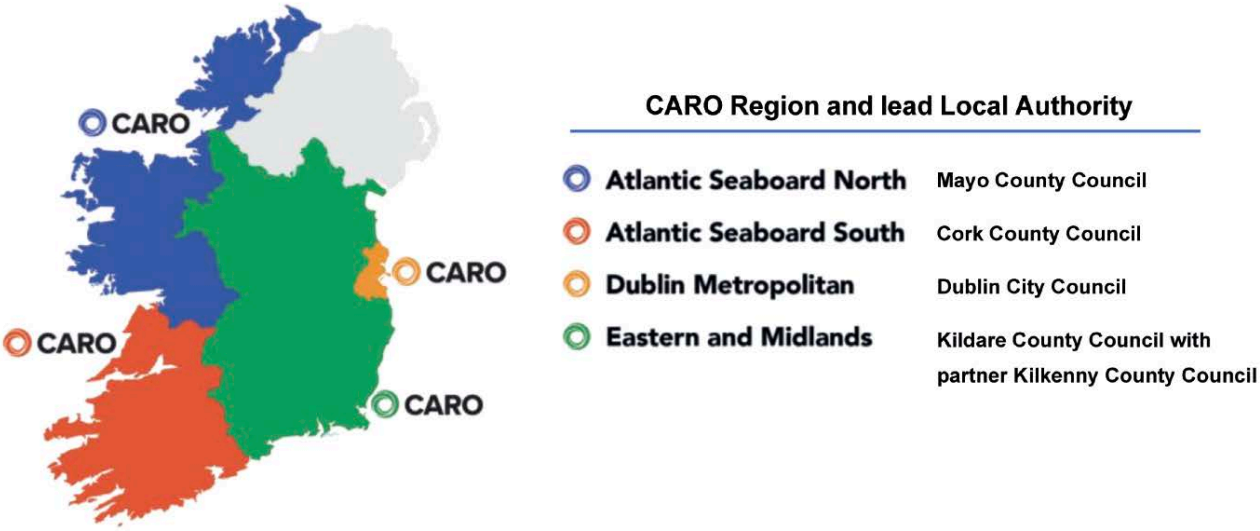
8.4.1 Climate Action Regional Offices

In accordance with the National Adaptation Framework, four Local Authority Climate Action Regional Offices (CAROs) were established in 2018. Their role was originally to

support local authorities in the preparation of their local adaptation strategies, but has since expanded to include supporting coordination among local authorities and other stakeholders and the development of regional specialisms on climate issues. With the support of the CAROs, all 31 local authorities developed and adopted climate change adaptation strategies in 2019. This entailed a particular focus on mainstreaming climate change considerations into sectoral plans, policies and operations of the local authority, along with training and capacity building on climate issues within the sector.

There are four CAROs, namely: Atlantic Seaboard North, Atlantic Seaboard South, Dublin Metropolitan, and Eastern and Midlands. The designated regional specialisms for each are as follows:

- Atlantic Seaboard North – coastal erosion and flooding, Atlantic storms and groundwater flooding.
- Atlantic Seaboard South – sea level rise, coastal erosion and flooding.
- Eastern and Midlands Regional – fluvial and pluvial flooding.
- Dublin Metropolitan area – urban heat effect, urban freezing and urban pluvial flooding.



Source: Climate Action Regional Office

8.4.2 Local Government Initiatives

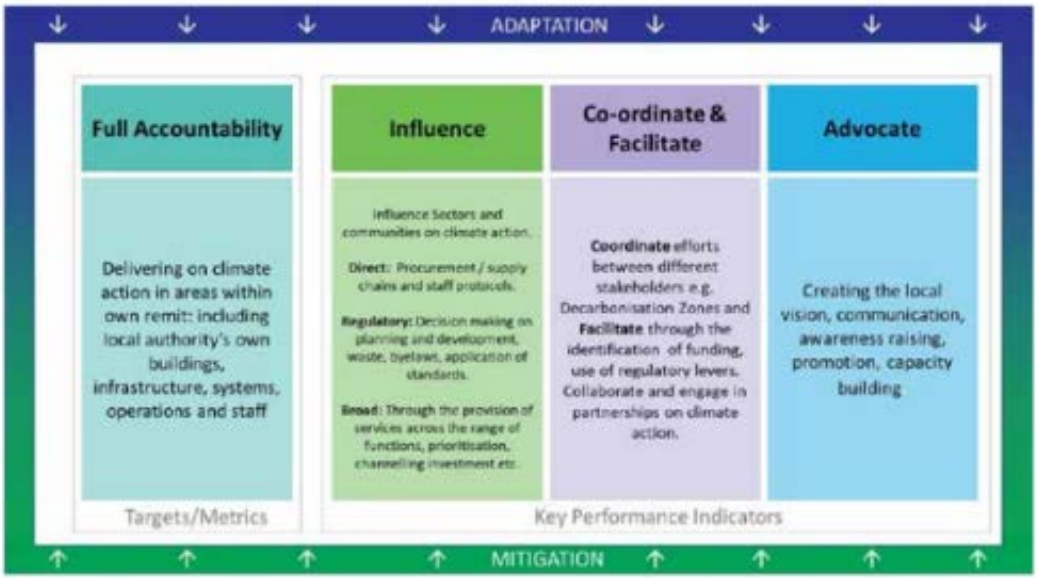
In 2020, the City and County Management Association set out a series of high level strategic goals to deliver transformative change and measurable climate action at local level. These included:

- Fostering governance, leadership, and partnerships
- Achieving medium and long-term carbon emission and energy efficiency targets
- Delivering on climate adaptation and resilience

- Mobilising local communities
- Supporting the transition to an inclusive, net zero and circular economy
- Achieving a 'just transition' particularly for local areas where decarbonisation is likely to have negative economic implications.

This approach implies that local authorities will have to build upon their track record of working with multiple stakeholders at local level to address complex challenges, advocating, influencing and working with key partners in their local area. It also implies the need for local authorities to 'lead by example' in reducing their own emissions and tackling emissions levels arising from their own in-house operations. Many local authorities have established cross departmental, multi-disciplinary climate action teams comprised of staff with a range of expertise. These teams are tasked with coordinating climate action activities across local authority structures and working with the CAROs in the coordination of activities across the region as well as implementing a range of actions.

Table 8.1 Local Authority Scope on Climate Action



Source: Local Authority Climate Action Plan Guidelines

Specific measures in this respect have included:

- Energy management and energy efficiency initiatives
- 'Green procurement' and integrating environmental criteria in tendering processes
- Mitigation measures undertaken by local authorities in a range of areas such as fleet management, retrofitting, public lighting, active travel/ public realm, facilities/ depots etc.
- The 'mainstreaming' of climate action considerations in its various plans, programmes and policies.

It is clear that this is an evolving area of local government in supporting sustainability and responding to climate change at local level and can be expected to develop further into the future. Key challenges in this area are likely to include:

- Local adaptation and mitigation measures
- Connecting with local communities
- Facilitating more active and sustainable transport
- Spatial planning
- Taking advantage of opportunities
- Managing the 'just transition'

Adaptation and Mitigation

As noted above, this refers to the need to build resilience, particularly at local level, to the foreseeable consequences of climate change, including those identified in local climate action plans. There will also be an obligation on all sectors, including local government, to significantly improve energy performance and to reduce carbon footprints. Revisions to the Energy Performance of Buildings Directive require dramatic improvements in the energy efficiency of local authority buildings. Significant areas of work in the area of climate change adaptation thus far have included work undertaken as part of the Catchment-based Flood Risk Assessment and Management (CFRAM) programme, and specific projects such as bridge rehabilitation.

Connecting with local communities

Local communities are a significant user of energy, but equally there is also increased interest in issues of sustainability, particularly on the part of the younger generation. Future work may entail increased interaction between staff, elected members and local communities to meet the challenges which lie ahead, in some cases drawing on existing structures such as Municipal Districts, the Public Participation Network (PPN), other fora (such as Tidy Towns) as well as other local development or community groups to pilot or experiment with new initiatives.

One initiative in this area has been the Community Climate Action Programme, which has provided financial support to local authorities working with community groups on projects designed to support the move to a low carbon community, such as developing community gardens and community composting, installing EV chargers or bike rack facilities, energy projects in areas such as lighting upgrades or solar energy, and water refill stations and rainwater harvesting.

Similarly, the Sustainable Energy Communities (SECs) Programme, supported by local authorities and the Sustainable Energy Authority Ireland, and which supported local communities in working together to develop a local sustainable energy system drawing on smart and renewable energy systems. Sustainable Energy Communities involve a bottom up approach to energy management at local level.

Another example includes the GAA Green Club Programme, a partnership between the GAA and local authorities. The programme is structured around different thematic areas, including: energy; water; waste; biodiversity; and travel/transport, and involves local authorities working with participating clubs to assist them in delivering actions under each area. The idea here is that through promoting these changes within GAA clubs, this will hopefully have a wide effect and prompt further similar initiatives within the broader community.

Active Travel

The National Development Plan envisages a significant shift towards the use of public transport, cycling, walking and trip reduction as well as improved access to transport services. With financial assistance from the National Transport Authority, local authorities have established active travel teams to construct and improve infrastructure for these transport modes, including dedicated cycleways, greenways, improved walking infrastructure and other efforts to make both walking and cycling safer and more attractive at local level (see Lesson on Roads and Transportation for further details).

Spatial Planning

There is also a significant long-term link between spatial planning and climate action/ sustainability. This will have a particular emphasis on integrating climate action into development plan objectives – including addressing some sensitive topics such as settlement patterns, compact urban forms, promoting and facilitating active travel, along with policies relating to renewable energy etc. It may also include nature based initiatives, enhanced green infrastructure and protecting biodiversity. Planning decisions also involve identifying 'decarbonising zones' within local authority areas where a more intensive set of specific climate mitigation, adaptation and biodiversity measures are to be implemented by different actors in a targeted manner

Opportunities

Climate action is not entirely about risk, and there are significant opportunities for businesses and local economies, including new products, services, technologies, educational streams, research and development, energy systems etc. This greening of economies is a long-term trend, but a range of economic opportunities have been identified with Local Enterprise Offices (LEOs) playing a role in providing training and supports for local businesses as the country transitions to a net-zero carbon economy.

Just Transition

As part of the European 'Green Deal', the EU has put in place a Just Transition Fund to provide support to areas facing serious social and economic challenges due to the transition towards climate neutrality. At national level, a number of initiatives to address Just Transition have been developed, such as the setting up of the Midlands Regional Transition Team (MRTT) and the establishment by the Government of a Just Transition

Fund as well as an Office of a Just Transition Commissioner. The MRTT was established by Offaly County Council in response to Bord na Mona's decarbonising programme involving the closure of several bogs as well as two power stations in Offaly and Longford. The programme seeks to draw on funding opportunities to support those losing jobs in the peat industry and to attract investment and secure alternative employment opportunities in the region.

The MRTT was subsequently restructured to reflect the wider Midlands area and established key working groups dealing with education and training, employment generation and pathways, and comm unity assistance. All local authorities within the eight counties identified are represented along with members of the Oireachtas, Regional Assemblies, the Just Transition Commissioner, Bord na Mona, ESB and trade union representatives.

8.5 CONCLUSION

Clearly, local government's role in this area can be expected to develop into the future as the broader context for climate change evolves. From 2022, local authorities started published key performance indicators in areas such as active travel initiatives, housing retrofits, greenhouse gas reductions, responding to emergency weather events, and implementing climate adaptation initiatives.

As emphasised at the outset, while there are many examples of local authority projects and initiatives that clearly fall under the banner of 'climate action', climate action transcends several other service areas, from spatial planning and sustainable development patterns, emergency planning, biodiversity and environmental protection, promoting and facilitating active and more sustainable modes of transport, building control regulations enforcement, to retrofitting the local authorities' own housing stock and other local authority buildings. In this respect climate action cannot be seen as the responsibility of a single section or group within the local authority, but rather something that touches on all service areas.

KEY CONCEPTS

- Sustainable Development
- Sustainable Development Goals
- United Nations Framework Convention on Climate Change (UNFCCC)
- Paris Agreement
- European Green Deal
- Mitigation
- Adaptation
- Climate Action Plan
- Local Climate Change Adaptation Strategy
- Local Authority Climate Action Plan

- Local Authority Climate Action Regional Office (CARO)
- Decarbonising Zones
- Just Transition

STUDENT ACTIVITIES

1. Consult the local authority climate action plan for your local area. How well equipped do you think your area is to cope with some of the possible effects of climate change?

2. What actions can local authorities undertake on their own initiative in this area? What actions require extensive interaction with local communities, or other stakeholders such as other public bodies or the private and voluntary sector?

3. Reflect on the service area in which you currently work. What do you consider to be the top three implications of climate change for this area of work?